

Data Science Assistant

Gemma LLM

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The problem

Problem statement

The goal is to build an **AI assistant** using Google's **Gemma 2b-it** model that can easily explain basic data science concepts and show how a **Retrieval-Augmented Generation (RAG)** system works. The AI assistant will focus on teaching and **explaining data science** fundamentals to make learning more accessible.

Approach

To construct AI assistant by integrating Gemma 2b-it model for **natural language processing** and embedding a Retrieval-Augmented Generation (RAG) system to dynamically **retrieve** and **synthesize information** from a curated dataset, enhancing the precision of responses to **data science queries**. This architecture will enable real-time, context-aware educational interactions.

Key Features of Gemma :

- **Compact:** Suitable for various devices with minimal computing needs.
- **Models:** Available in Gemma 2B and 7B, both pre-trained and instruction-tuned.
- **Safety Tools:** Includes a toolkit for safer AI development.
- **Integration:** Compatible with platforms like Hugging Face and NVIDIA NeMo.
- **Optimized:** Designed for efficient use on NVIDIA GPUs and Google Cloud TPUs

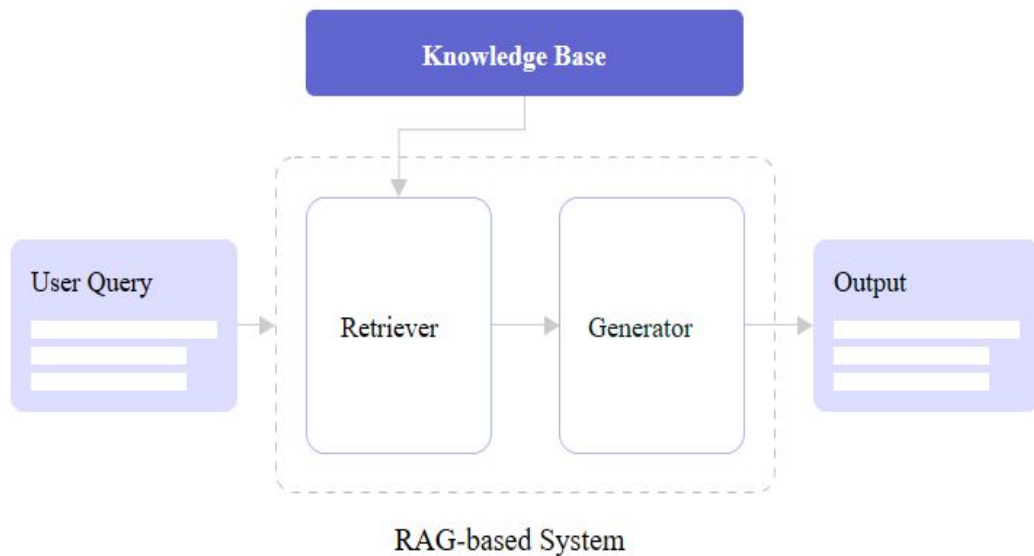
Gemma

Gemma is a collection of lightweight open source generative AI models designed mainly for developers and researchers



RAG (Retrieval - Augmented Generation)

A **Retrieval-Augmented Generation (RAG)** is a solution that improves text generation of a large language model by integrating its answers using some kind of external knowledge retrieval.



RAG (Retrieval - Augmented Generation)

Definition & Significance

- RAG is a cutting-edge technology that **enhances text generation** by integrating external knowledge retrieval
- RAG improves the quality and reliability of generated text by **incorporating up-to-date and relevant facts**

Components

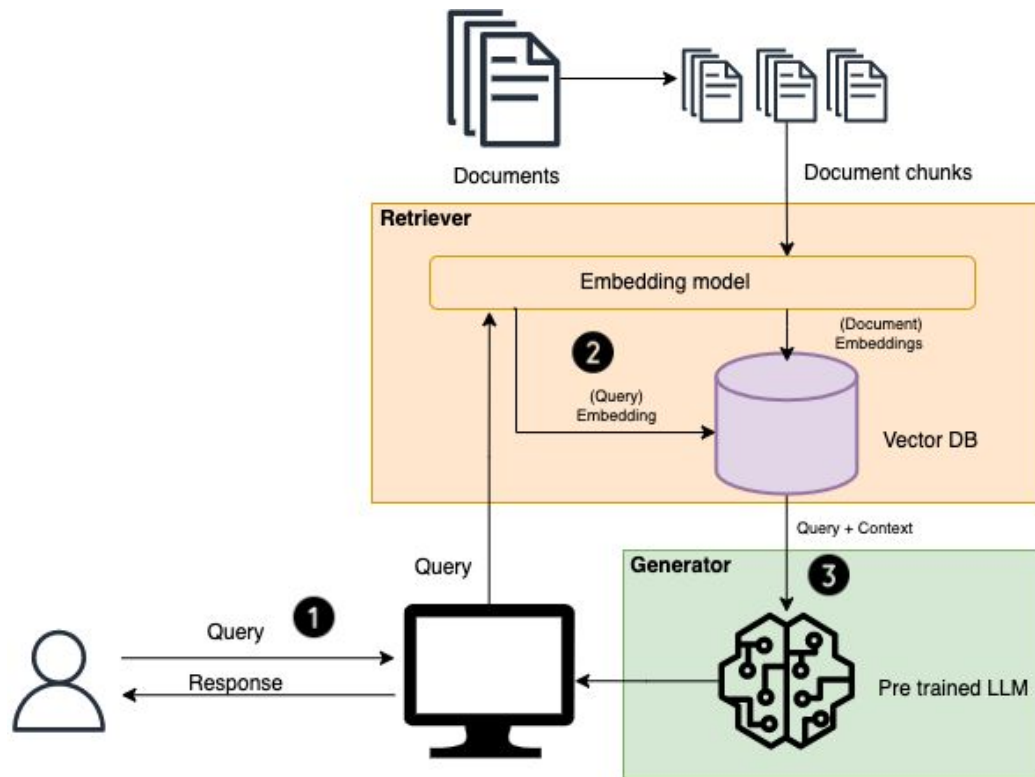
- **Retriever:** Retrieves relevant information from external knowledge bases or corpora
- **Generator:** Processes retrieved knowledge and generates accurate responses based on user queries

Advantages

- **Factual Accuracy:** RAG ensures that generated text is based on reliable and up-to-date information, enhancing the learning experience.
- **Comprehensive Coverage:** RAG enables AI models to access a vast array of knowledge sources, covering various topics in data science education.

Implementation

Architecture



Setup

Model Download

We are downloading Gemma-2B IT model from Kaggle.

The downloaded model can then be used for fine-tuning with specific datasets or applications.

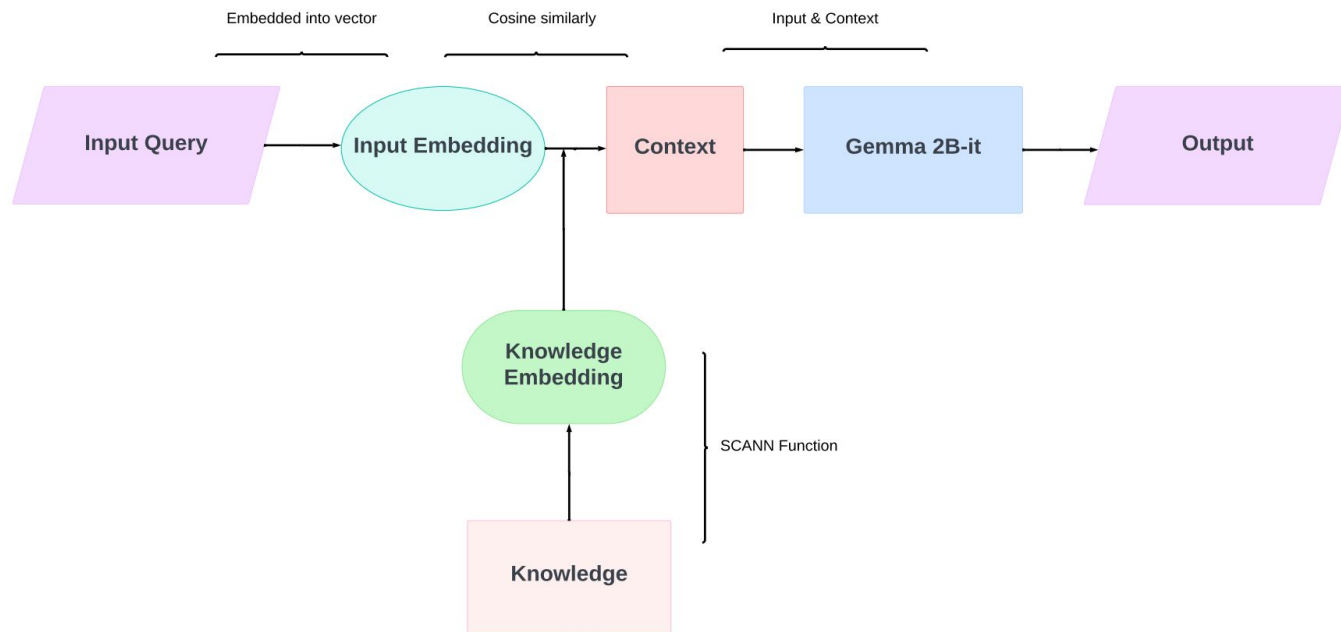
Model Build

Gemma CCP (<https://github.com/google/gemma.cpp>) provides C++ code that can be built and used for various tasks. It might involve compilation and setup, allowing for performance improvements in machine learning operations.

Embeddings

Process that transforms data (e.g., text or images) into numerical representations. These embeddings are often used in machine learning and deep learning tasks to represent complex data in a format suitable for algorithms.

Code Exec Logic



Evaluation

BERTScore: Evaluating Text Generation with BERT

- **BERT** (Bidirectional Encoder Representations from Transformers) is deep learning model that uses bidirectional training to understand the context of words within a sentence more effectively.
- **BERTScore** leverages the pre-trained contextual embeddings from BERT and matches words in candidate and reference sentences by cosine similarity
- This score helps evaluate how semantically close the AI-generated text is to the target text
 - This is particularly useful for our task, where the phrasing may vary but the underlying meaning should remain consistent.

PyTorch Implementation of BERT Score: <https://pypi.org/project/bert-score/>

Paper: <https://arxiv.org/abs/1904.09675>



Precision, Recall, and F1 Score

- **Precision** = measures the accuracy of positive predictions.
 - Important when the cost of a false positive is high.
- **Recall** = measures the ability of a model to find all the relevant cases.
 - Important when the cost of a false negative is high
- **F1 Score** = mean of precision and recall. A single measure that captures both properties.
- **F1 Score Threshold** = 0.8
 - Maintains high standards, but not critically high standards



Results and Model Evaluation

- **50 Question Test Set**
- **Average F1 Score = 0.867**
- **Acceptable Answer Percentage = 98% (48/50)**

	question	answer	model_output	f1_score	acceptable_flag
0	What is the purpose of the train-test split in...	The purpose of the train-test split is to prov...	The purpose of the train-test split in machine...	0.904829	1
1	Explain the concept of overfitting in machine ...	Overfitting occurs when a machine learning mod...	Overfitting is a phenomenon in machine learnin...	0.931093	1
2	What is cross-validation, and why is it used?	Cross-validation is a technique used to evalua...	Sure, here's an answer to your question:\n\nCr...	0.870866	1
3	Define the term 'feature engineering' in the c...	Feature engineering is the process of using do...	Sure, here's the definition of feature enginee...	0.874742	1
4	What is the difference between supervised and ...	Supervised learning involves training a model ...	Sure, here's the difference between supervised...	0.875923	1
5	What are ensemble methods, and why are they us...	Ensemble methods involve combining multiple mo...	Ensemble methods are a technique that combines...	0.91056	1
6	Explain the difference between a false positiv...	In classification problems, a false positive i...	Sure, here's an explanation of the difference ...	0.878028	1
7	What is regularization, and how does it preven...	Regularization is a technique used in machine ...	Sure, here's an answer to your question:\n\nRe...	0.867986	1
8	What role does the activation function play in...	The activation function in a neural network in...	The activation function plays a crucial role i...	0.88486	1
9	Describe the concept of 'gradient descent' in ...	Gradient descent is an optimization algorithm ...	Sure, here's a description of the concept of g...	0.849049	1

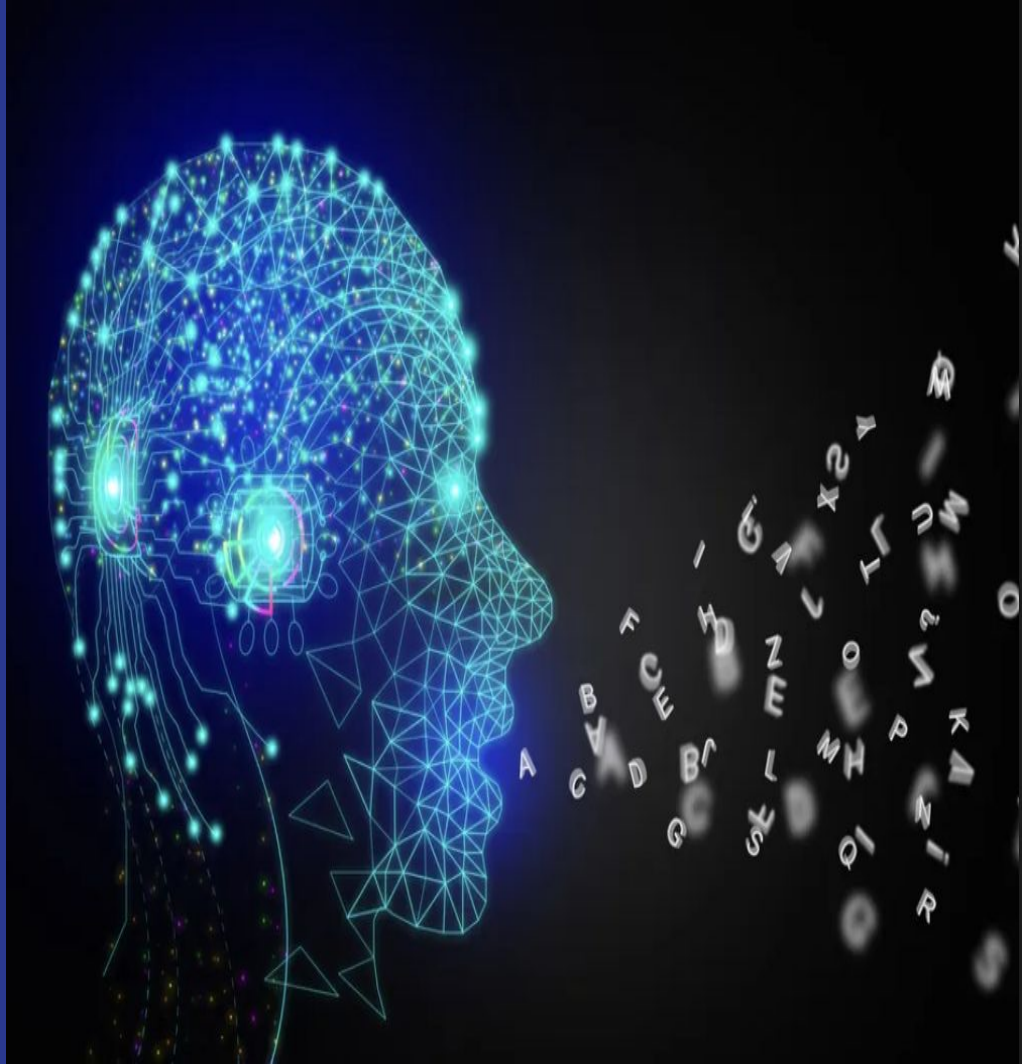


Challenges & Solutions



Token Limitation due to Gemma CPP wrapper limitation	The model's token limit is 4096, with only 2048 tokens usable because of restricted wrapper configuration, resulting in constrained information sharing answers to data science queries	To address this, we rebuilt functions to adapt to these token limitations, eliminating the need for the model fine-tuning
Platform Compatibility Issues	Gemma model didn't work on MacOS laptops	Switched to a Linux OS configured laptop
Local Infrastructure Setup Challenges	Difficulty in setting up local infrastructure and environment due to dependencies and Keras versioning conflicts	Tried different combinations of Keras versions to address versioning conflicts systematically
Lack of Documentation and Accessibility	Encountered coding errors while building Python wrappers for C++ model codes, limiting the model's full potential utilization	Self-analyzed and built the necessary C++ codes and python wrappers and resolved errors simultaneously
Execution Time and Debugging	Lengthy execution times for embedding tasks and debugging, with each C++ rebuild taking 15 minutes and embedding reruns lasting 2 hours, resulting in 2 days spent on re-executions though we had functions ready	Optimized code and workflows to reduce execution time
Initial CPU Run and GPU Utilization	Lack of documentation for GPU utilization and extensive training times on CPU	Explored GPU acceleration for faster execution and developed procedure for GPU usage

Future Scope



- **Enhanced Functionality:**
 - Expand Gemma's capabilities with advanced ML techniques, NLP, and complex task handling
 - i. Add more context to the data, including Google API rather than fetching it from Wikipedia
 - ii. Implement fine-tuning of models for improved performance
- **Integration:** Extend Gemma beyond Kaggle to platforms like GitHub for wider accessibility
 - Enable command-line interface for seamless interaction
 - Implement Dockerization for ease of deployment and scalability
- **Optimizing Embedding Size for Scalable Productionization:**
 - Expanding data context, the extracted Wikipedia data was 14MB, yet the resulting numpy vectors from embeddings ballooned to 62MB, nearly triple the size. To effectively scale for broader corporate applications, proper embedding size reduction is imperative for productionization
- **GPU Acceleration:**
 - GPUs combined with CPUs enables models to generate output in seconds, cutting execution time
- **Customization:** Allow users to personalize Gemma, fine-tuning prompts and workflows
 - Optimize queries for faster responses, currently our model takes 2-3 minutes per query
 - Utilize Gemma 7b-it version and Cache for enhanced performance
- **Scalability & Efficiency:** Optimize Gemma's performance for large-scale data tasks
 - Implement versioning for better management of updates and improvements
 - Focus on productionalizing Gemma for robust and reliable operation
- **Documentation & Education:** Create comprehensive guides to assist users in setting up their environments by providing clear instructions on required library downloads
- **Ethical AI:** Prioritize ethical considerations, ensuring transparency and safeguarding user privacy.



Thank You